

Interior Point Method - Proof Outline

The basic formulation Barrier function: $F_0(x)$

For LP: " $Ax \leq b$ ", $F_0(x) = -\sum_{i=1}^m \log(b_i - a_i^T x)$

$$F_t(x) = t \cdot c^T x + F_0(x)$$

$$x_t \leftarrow \operatorname{argmin} : F_t(x)$$

Three key objectives

step 1: We can optimize F_t quickly via
Newton's method

step 2: We can increase t quickly: $t^+ = (1+\alpha)t$
while staying in quad conv. regime:

$x_t = \operatorname{argmin} F_t$. We need x_t to be in quad conv regime
of $F_{t^+}(x)$

step 3: $(c^T x_t - c^T x^*) \rightarrow 0$ quickly.

Key Definitions

Local norm:

$$\|h\|_x^2 \triangleq \|h\|_{\nabla^2 F_0}^2 \doteq h^T \nabla^2 F_0 h$$

$$\nabla^2 F_t = \nabla^2 F_0 \quad \forall t$$

A function f is M -self-concordant if

$$\|\nabla^3 f(x)[u, u, u]\| \leq 2 \cdot M \cdot \|h\|_x^3$$

Step 1: Quadratic Convergence

For Newton, recall that we showed we were in the quadratic convergence regime as soon as we had $\|\nabla f(x)\|_2$ small.

We replace this with the local norm:

$$\|\nabla f(x)\|_x^*$$

← dual norm, b/c it's the derivative (recall Mirror Descent)

Def'n: $\lambda_f(x) \triangleq \|\nabla f(x)\|_x^*$ - "Newton Decrement"

Step 1: Quadratic Convergence - cont -

Lemma:

$$\text{If } \lambda_f(x) < 1$$

$$\Rightarrow \|x - x^*\|_x \leq \frac{\lambda_f(x)}{1 - \lambda_f(x)}$$

Step 1: Quadratic Convergence - cont -

Theorem: If $\lambda_f(x) \leq 1/4$, then

$$\text{for } x^+ = x - (\nabla^2 f(x))^{-1} \nabla f(x)$$

$$\underline{\text{then}} : \lambda_f(x^+) \leq 2 \lambda_f(x)^2$$

$$\Rightarrow \lambda_f(x_{k+1}) \leq 2^{2^k} \lambda_f(x_1)^{2^k}$$

$$\text{if } \lambda_f(x_1) \leq 1/4$$

$$\Rightarrow \lambda_f(x_{k+1}) \leq (1/2)^{2^k}$$

$$\text{Now using Lemma : } \|x_{k+1} - x^*\|_2 \leq \frac{\lambda_f(x_{k+1})}{1 - \lambda_f(x_{k+1})}$$

Reference :

Nesterov's "Lectures on
Convex Optimization"